

Residual Limits of a Load-Normalized Spine-Neck Ratio in Compartmental Models

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Abstract

Dendritic spine-neck resistance is interpretable only relative to the dendritic load it drives. We define $SMI = R_{\text{neck}}/R_{\text{in},d}$, where $R_{\text{in},d}$ is the spine-omitted dendrite-soma input resistance, yielding the DC divider expectation $\Gamma_{\text{div}} = 1/(1 + SMI)$. Using SPINE, a transparent Python compartmental implementation, we analyzed residuals between conductance-based peak local transfer and this divider. Across 3718 rows, residuals had median absolute value 0.0548 and maximum 0.492, and were usually negative. Descriptor comparisons showed that SMI is a compact local-transfer coordinate, whereas residuals, head amplitude, and somatic transfer require component resistances, impedance/dynamic descriptors, synaptic conductance scale, active state, and downstream filtering. Deterministic uncertainty stabilized sampled low/intermediate behavior but contained zero high-SMI rows. Independent direct-matrix and DC analytic benchmarks reproduced passive reference behavior to roundoff. SPINE therefore frames spine-neck isolation as an expected divider plus residuals that mark limits of a scalar descriptor.

Keywords: dendritic spines; compartmental modeling; cable theory; voltage divider; electrical isolation; impedance; reproducible modeling

Dendritic spines place excitatory synapses in small postsynaptic compartments connected to the parent dendrite by narrow necks. These necks can impede current flow, alter local voltage, and separate spine-head depolarization from parent-dendrite or somatic impact [1–9]. A neck resistance by itself, however, is incomplete: the same series resistance has different consequences depending on the load presented by the dendritic and somatic circuit.

The first-order electrical expectation is the classical voltage divider. If a spine-head voltage drives a series neck resistance R_{neck} into a dendrite-soma load with input resistance $R_{\text{in},d}$, the DC small-signal local transfer is

$$\Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + R_{\text{neck}}/R_{\text{in},d}}. \quad (1)$$

For compact notation, we denote the author-defined ratio $R_{\text{neck}}/R_{\text{in},d}$ as SMI. The abbreviation is used as a manuscript and software shorthand for a load-normalized spine-neck ratio, not as a claim that the ratio is a field-standard morphology index or a new cable-theory law.

This distinction matters because the divider expectation is not the same as a transient synaptic peak response. Conductance-based excitation changes driving force while the response evolves; capacitance, local impedance, dendrite-to-soma filtering, active conductances, morphology, and measurement uncertainty can all alter observed voltages [10–14]. A scalar ratio can be useful when the target is local low-frequency isolation, yet fail for head amplitude, somatic voltage, nonlinear response, or categorical class assignment.

The present study therefore reframes SPINE around the residual from the analytic divider:

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (2)$$

The ratio defines the expected local divider coordinate; the residual tests where the full simulated response leaves that coordinate. This makes negative results central rather than embarrassing. Large residuals, equal-ratio nonequivalence, and amplitude failures identify the electrical variables that the compact ratio cannot contain.

We ask five bounded questions. First, do the pre-specified low-, intermediate-, and high-isolation reference cases reproduce the intended passive responses? Second, how far do observed peak local transfers depart from Γ_{div} across existing passive, active, uncertainty, and exploratory rows? Third, when is the ratio more useful than its component resistances, and when do component, impedance, dynamic, or conductance descriptors carry additional information? Fourth, how should deterministic uncertainty and radius-error screens be reported without implying population inference? Fifth, do independent matrix and analytic benchmarks support the computational credibility of the from-scratch passive implementation? The resulting thesis is deliberately scoped: the load-normalized ratio recovers the classical local divider expectation, and SPINE maps the residual domains where a single scalar is insufficient.

1 Results

1.1 Reference cases reproduce the intended passive ordering

The baseline reference configuration generated the intended progression from near-isopotential head-dendrite responses to strong local isolation (Figures 1 and 2). The simulations used the baseline parameters in Table 1, $\Delta t = 0.01$ ms, a single head synapse at 20 ms, and the predefined 50 ms metric window. All nine low-, intermediate-, and high-isolation target values passed the predefined tolerance rule.

The low-SMI case produced $\text{SMI} = 0.00881$, $\Gamma_{h \rightarrow d} = 0.984$, $\Gamma_{h \rightarrow s} = 0.357$, and $A_h = 1.82$ mV. The intermediate case produced $\text{SMI} = 0.115$, $\Gamma_{h \rightarrow d} = 0.751$, $\Gamma_{h \rightarrow s} = 0.273$, and $A_h = 2.36$ mV. The high-SMI case produced $\text{SMI} = 1.32$, $\Gamma_{h \rightarrow d} = 0.102$, $\Gamma_{h \rightarrow s} = 0.0376$, and $A_h = 15.1$ mV. These target cases support implementation consistency for the reference equations, units, synaptic normalization, solver, and metric definitions in the compact passive architecture; they do not calibrate the model to a specific biological neuron.

The same traces show why the divider residual is needed. The baseline divider predictions were 0.991, 0.897, and 0.431 for the low, intermediate, and high cases. The corresponding residuals $\Gamma_{h \rightarrow d} - \Gamma_{\text{div}}$ were -0.00729, -0.1458, and -0.3286. The low case was close to the DC divider, whereas the intermediate and high transient peak ratios fell well below it.

1.2 The analytic divider is a reference, not the peak-response result

Across 3718 existing residual rows, the median absolute residual from the divider was 0.0548, the mean absolute residual was 0.0734, and the root-mean-square residual was 0.103. The maximum

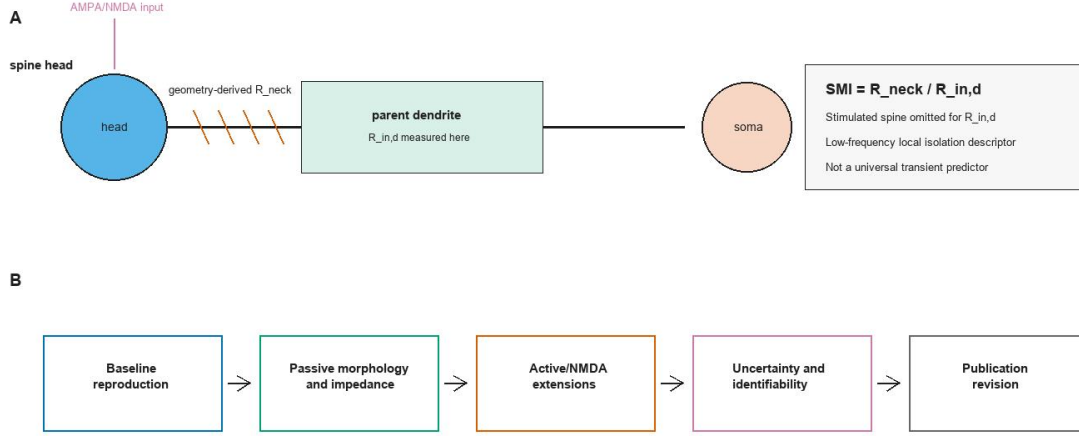


Figure 1: SPINE reference architecture and residual workflow. A conductance-based excitatory input drives an explicit spine head coupled to the parent dendrite through a geometry-derived neck resistance. The dendrite-soma input resistance is measured after omitting the stimulated spine, yielding $SMI = R_{neck}/R_{in,d}$ and the analytic local divider expectation $\Gamma_{div} = 1/(1 + SMI)$.

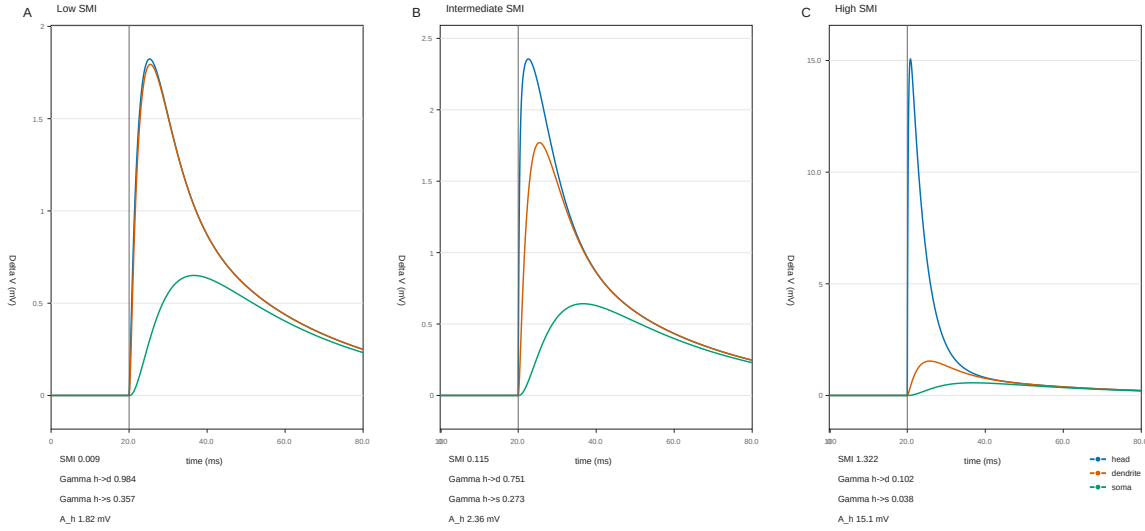


Figure 2: Baseline passive reference traces. Low-, intermediate-, and high-SMI traces show the intended local coupling progression. The reference cases also demonstrate that target-case ordering does not imply exact equality to the DC divider: residuals are small for the low case and larger for intermediate/high transient peak responses.

absolute residual was 0.492. Residuals were negative in 3680 rows and positive in 38 rows, so the DC divider usually overpredicted the observed conductance-based peak local transfer (Figure 3).

The largest departures were scientifically informative rather than numerical defects. The matched-neck heterogeneous-load sweep produced the largest absolute residual, with $\text{SMI} = 0.452$, observed $\Gamma_{h \rightarrow d} = 0.196$, divider prediction 0.689, and residual -0.492. Fixed-load high-SMI rows also showed large relative departures. These patterns are consistent with the distinction between a low-frequency small-signal divider and a transient conductance-driven peak voltage ratio.

Excluding exploratory rows left the same conclusion: 3584 non-exploratory rows had median absolute residual 0.0563, mean absolute residual 0.0748, and only two positive residual rows. The divider relation therefore remains the correct analytic baseline, but the residual field is the empirical domain-of-validity map.

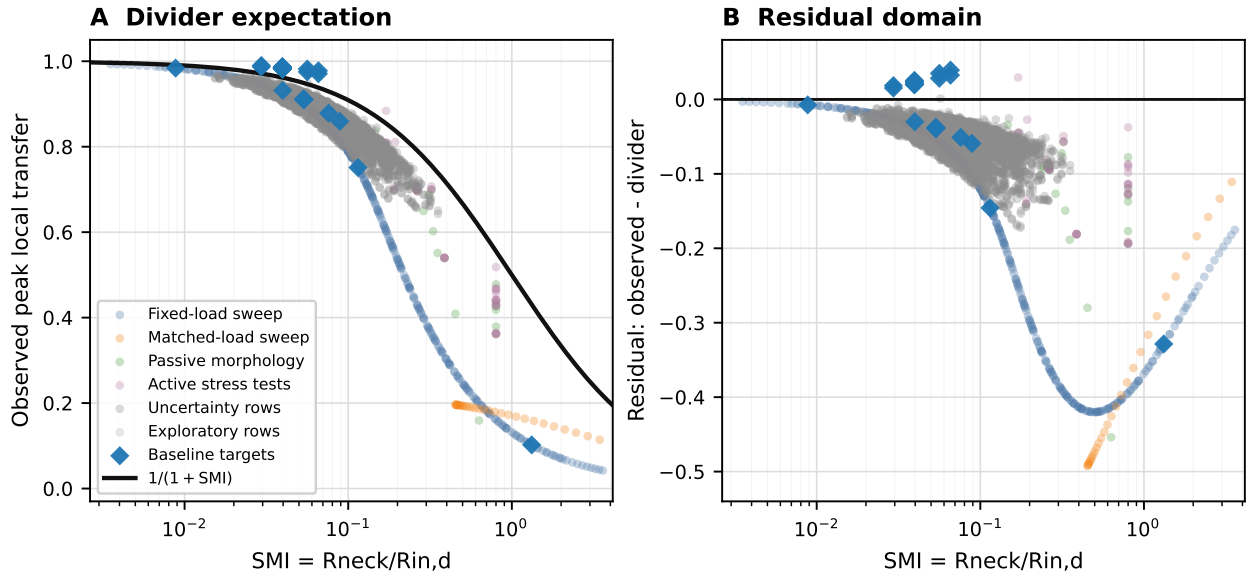


Figure 3: Analytic divider overlay and residual domain. Observed peak local transfer is compared with the DC divider expectation $\Gamma_{\text{div}} = 1/(1 + \text{SMI})$ across 3718 derived rows. Most points fall below the divider curve, and the residual panel makes the departure visible. Baseline targets are highlighted; exploratory rows are included only as separated stress-test context.

1.3 Load normalization is useful but compact

The fixed-load geometry sweep behaved as a constructional sanity check. Longer and narrower necks increased R_{neck} , increased head depolarization, and decreased downstream transfer. Because $R_{\text{in},d}$ was fixed, SMI and R_{neck} had identical rank ordering. This result should not be read as evidence that the ratio adds information under fixed-load conditions.

The matched-neck experiment held $R_{\text{neck}} = 100 \text{ M}\Omega$ while varying dendritic leak from 0.1 to 30 nS (Figure 4). Under this manipulation, neck resistance alone could not rank the cases because there was only one neck-resistance value. Changing the dendritic load changed $R_{\text{in},d}$, and therefore changed SMI and downstream transfer. The load term is therefore mechanistically meaningful, but only within the target it was designed to represent.

Phase 2 descriptor comparisons made this compactness explicit. For non-exploratory local transfer, SMI and Γ_{div} had the same absolute Spearman association because the divider is a monotone transform of the ratio, but the analytic divider was the better direct scalar prediction (cross-validated RMSE 0.052 versus 0.091 for raw SMI). The log component pair improved on raw SMI but did

not outperform the analytic divider for first-order local transfer. Thus the ratio is useful as the coordinate defining the divider, not as a universally superior scalar.

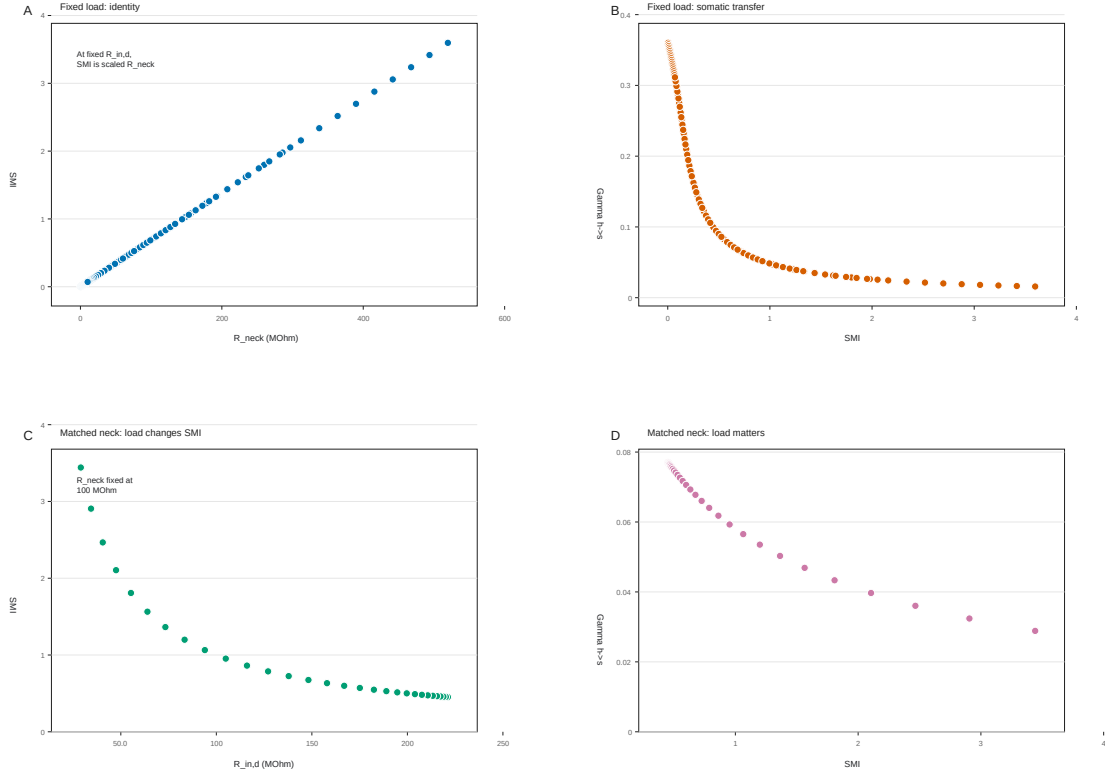


Figure 4: Fixed-load and matched-neck controls. At fixed dendritic input resistance, SMI is a scaled neck-resistance coordinate. When R_{neck} is held fixed and dendritic load changes, the ratio changes with $R_{\text{in},d}$ and separates cases that neck resistance alone cannot order.

1.4 Residuals, amplitude, and somatic transfer require richer descriptors

The residual analysis separated local-divider usefulness from broader prediction. For signed and absolute divider residuals, dynamic SMI was the strongest available scalar by rank association (absolute Spearman 0.853), while R_{neck} exceeded raw SMI (about 0.765 versus 0.704). In rows with synaptic conductance scale, conductance plus log-component models sharply reduced residual prediction error. Residuals are therefore better treated as a dynamic/conductance/impedance problem than as evidence that the DC ratio alone is sufficient.

Head amplitude was still more target-specific. Across non-exploratory rows, synaptic conductance scale had absolute Spearman association 0.982 with A_h , whereas raw SMI was only 0.143. Somatic transfer depended on downstream filtering: transfer gain and transfer impedance exceeded raw SMI for $\Gamma_{h \rightarrow s}$ in the Phase 2 summary. These results support a target-specific descriptor interpretation: local divider reasoning, head amplitude, residuals, and somatic transfer require different descriptor families.

Designed passive counterexamples sharpened this interpretation. Passive iso-SMI cases with

different morphology produced relative spread 0.28 in $\Gamma_{h \rightarrow d}$ and 0.61 in A_h , both exceeding the predefined 20% failure threshold. Equal SMI is therefore not electrical equivalence, because the same ratio can hide different absolute resistances, capacitances, impedance spectra, synaptic drives, and downstream morphologies.

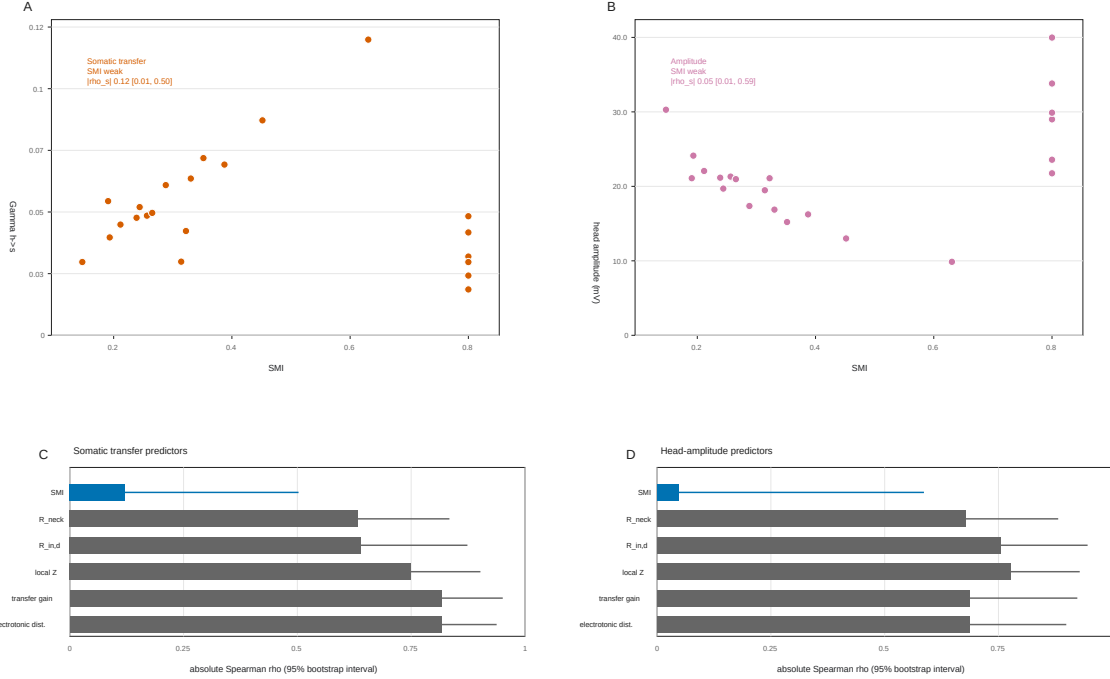


Figure 5: Passive limits of the compact ratio. Passive challenge rows show that local-divider information does not automatically transfer to head amplitude or somatic voltage. Component, impedance, transfer-gain, and synaptic-drive descriptors retain information that the ratio discards.

1.5 Active and NMDA stress tests preserve scope limits

Active validation passed all 12 predefined rows, including AMPA peak normalization, NMDA magnesium-block monotonicity, NMDA current-voltage relief, active-current signs, bounded gates, resting-state stability, timestep refinement, limiting-case solver comparison, strong-drive stability, and unit conversion checks. Active mechanisms were disabled in the reference baseline and enabled only in active-extension protocols.

In the designed active nonlinear suite, SMI remained strongly associated with local $\Gamma_{h \rightarrow d}$, but active mechanisms did not make the static ratio a general amplitude or somatic-transfer predictor (Figure 6). For active $\Gamma_{h \rightarrow s}$, input impedance at 50 Hz exceeded raw SMI (0.764 versus 0.339). For active A_h , frozen-gate transfer gain exceeded raw SMI (0.736 versus 0.236). AMPA+NMDA protocols with full restrained active conductances produced iso-SMI relative spread 0.86 in A_h and 0.35 in $\Gamma_{h \rightarrow d}$, exceeding the predefined active stress-test thresholds.

Thus active mechanisms do not overturn the divider framing; they make its scope more important. Voltage-dependent current and active state add response variables that are not contained in a static morphology/load ratio.

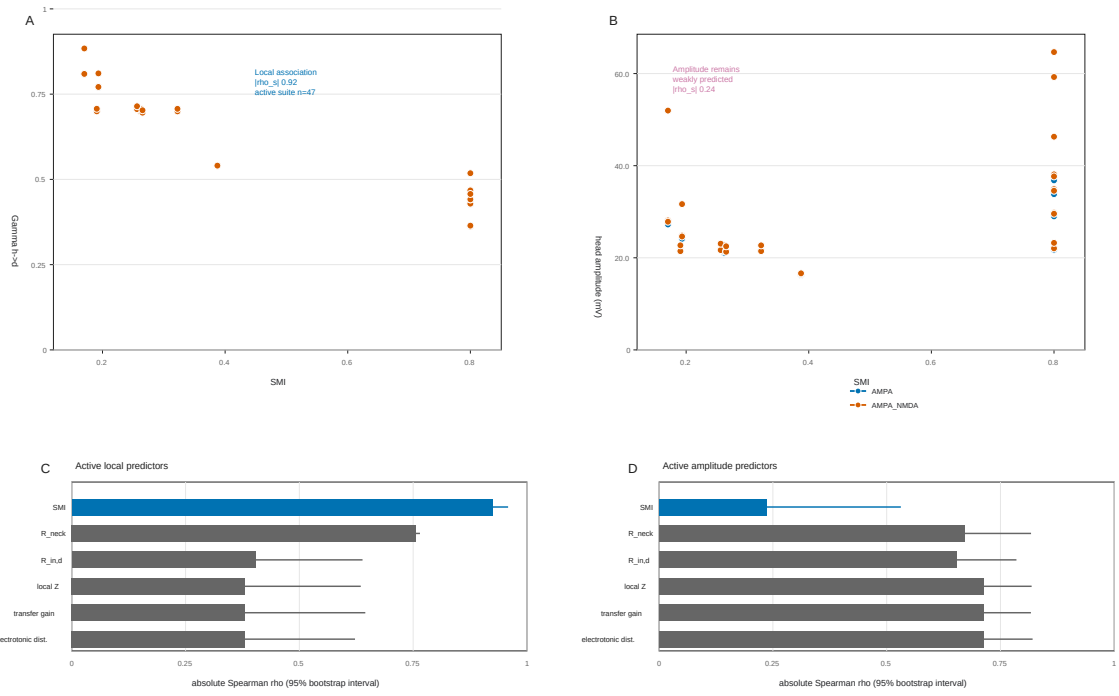


Figure 6: Active and NMDA stress-test boundaries. Active and NMDA-enabled protocols retain local ratio ordering in designed cases but increase nonequivalence for amplitude and other response targets. These rows are generic stress tests rather than cell-type-calibrated active reconstructions.

1.6 Deterministic uncertainty stabilizes sampled low/intermediate behavior but not high-SMI claims

Progressive uncertainty analysis resolved the small-sample instability seen in the original 48-versus-96 comparison. By $N=768$, final SMI median convergence passed: the 384-to-768 relative change was 0.012, below the predefined 0.10 threshold. The maximum final median relative change across tracked outputs was 0.029. Exact top-predictor ranking remained less stable than descriptor-family interpretation.

The $N=768$ ensemble does not support high-SMI uncertainty claims. It contained 757 low-SMI rows, 11 intermediate-SMI rows, and zero high-SMI rows. High-isolation conclusions therefore rest on baseline reference and designed challenge rows, not on the deterministic uncertainty ensemble.

Radius-only uncertainty produced 184 class flips out of 768 sampled parameter combinations, a design fraction of 24.0%. Low-class combinations flipped in 22.9% of cases, all 11 intermediate combinations flipped, and no high-SMI combinations were present. Because R_{neck} scales as $1/r_n^2$, continuous ratio values with explicit radius uncertainty are safer than treating low, intermediate, and high labels as sharply separated biological states.

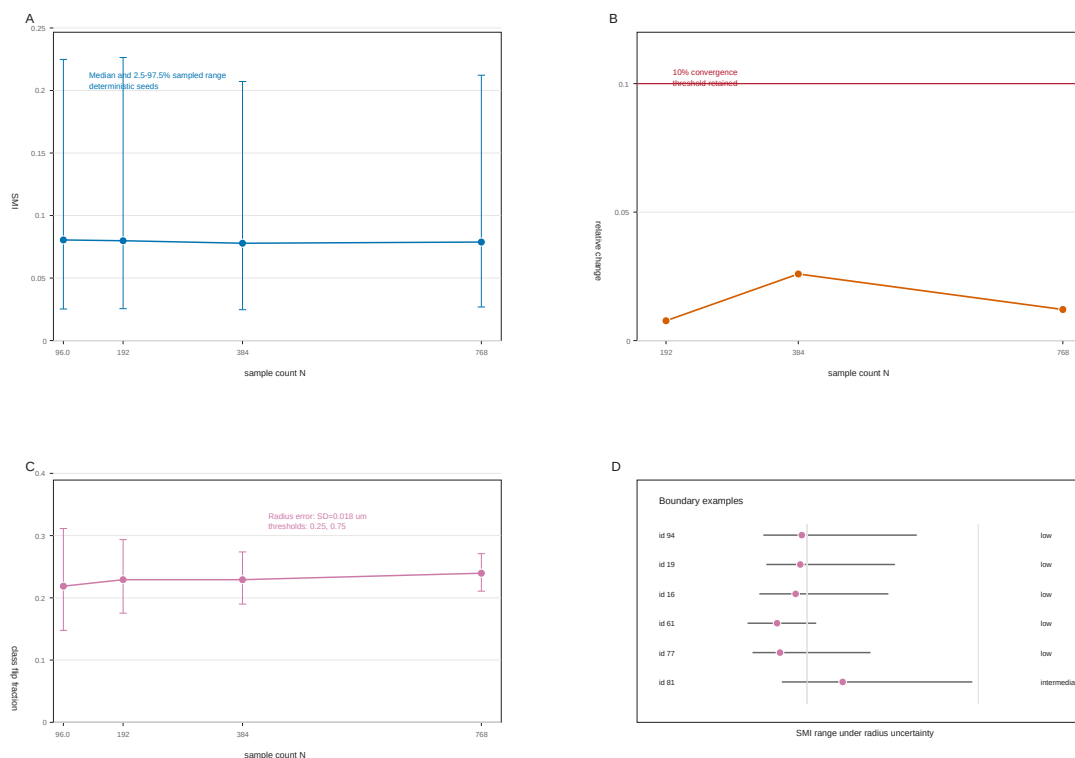


Figure 7: Deterministic uncertainty and radius sensitivity. Progressive sampling stabilized continuous low/intermediate SMI summaries by $N=768$, while radius-error propagation exposed threshold-class fragility. Fractions are design fractions over sampled parameter combinations, not biological rates.

1.7 Independent benchmarks support computational credibility within scope

Phase 4 addressed computational credibility beyond internal scheme agreement. An independently written direct-matrix implementation of the same passive three-compartment baseline reproduced existing SPINE baseline traces to numerical roundoff: the maximum all-trace absolute voltage difference was 1.25×10^{-13} mV, the maximum head-amplitude difference was 1.42×10^{-14} mV, and the maximum $\Gamma_{h \rightarrow d}$ difference was 7.55×10^{-15} . DC analytic checks reproduced the two-node spine-omitted $R_{in,d}$ value of 144.486692 M Ω to roundoff and confirmed the divider limits.

Backward Euler versus Crank-Nicolson peak differences were small in the baseline reference cases: maximum peak differences were 0.00203 mV for A_h , 0.000534 mV for A_d , 0.000189 mV for A_s , and 0.000500 for $\Gamma_{h \rightarrow d}$. These are internal numerical self-consistency checks. NEURON was unavailable in the execution environment, so no NEURON cross-check is claimed.

Earlier reviewer-access packages established manifest and checksum workflows. The finalization package supersedes those approval-gated drafts with an approved public-release staging repository and journal-specific submission bundle; authenticated GitHub upload and Zenodo DOI minting remain technical manual steps, not scientific or PI-approval blockers.

Table 1: Baseline reference passive parameters. Values are taken from the manuscript-faithful baseline configuration and the three pre-specified reference neck geometries.

Parameter	Meaning	Value
E_L	leak reversal	-70 mV
E_{syn}	excitatory reversal	0 mV
c_m	specific capacitance	0.01 pF μm^{-2}
$\bar{g}_L$	head specific leak	3×10^{-6} nS μm^{-2}
r_h	head radius	0.35 μm
C_d, C_s	dendrite and soma capacitance	120 pF, 220 pF
$g_{L,d}, g_{L,s}$	dendrite and soma leak	2.5 nS, 7.0 nS
g_{ds}	dendrite-soma coupling	12.0 nS
ρ_i	intracellular resistivity	100 Ω cm
g_{max}	peak synaptic conductance	1.4 nS
τ_r, τ_d	rise and decay constants	0.3 ms, 3.0 ms
$t_0, \Delta t$	event time and time step	20 ms, 0.01 ms
T	metric window	50 ms
$R_{in,d}$	spine-omitted dendrite-soma input resistance	144.487 M Ω
low neck	L_n, r_n, R_{neck}	0.25 μm , 0.25 μm , 1.273 M Ω
intermediate neck	L_n, r_n, R_{neck}	0.75 μm , 0.12 μm , 16.579 M Ω
high neck	L_n, r_n, R_{neck}	1.50 μm , 0.05 μm , 190.986 M Ω

2 Discussion

SPINE now supports a residual-domain interpretation of the load-normalized spine-neck ratio. The ratio $\text{SMI} = R_{neck}/R_{in,d}$ encodes the classical local divider expectation $\Gamma_{div} = 1/(1 + \text{SMI})$. The scientific work begins after that analytic reference is stated: transient conductance-based simulations usually produced peak local transfer below the DC divider, and the residuals identify where a single scalar loses information.

This reframing is important because it separates a true but limited electrical expectation from broader prediction. The low-SMI reference case was close to the divider limit, while intermediate

and high reference cases were substantially below the divider in peak transfer. Across the full derived table, residuals were usually negative and sometimes large. That pattern is not a failure to reproduce the divider; it is evidence that the divider is the low-frequency baseline and that peak synaptic responses live in a richer dynamic system.

The practical use pattern is therefore two-step. Use the load-normalized ratio to report first-pass local head-to-dendrite divider context. Then report residuals, component resistances, impedance, transfer gain, synaptic conductance scale, active state, morphology, and uncertainty whenever the target is peak transfer, head amplitude, somatic voltage, nonlinear response, class membership, or electrical equivalence.

The ratio succeeds locally because it compares a series neck resistance with the load driven by that resistance. This is the classical cable-theoretic intuition for spine-neck isolation [1, 2, 4, 10]. It also explains why the matched-neck control is meaningful: when R_{neck} is fixed, different dendritic loads produce different divider expectations.

The same logic explains why the ratio fails outside its target. Head amplitude depends on absolute synaptic conductance, driving-force reduction, local capacitance, and input impedance. Somatic transfer adds dendrite-to-soma filtering, a distinction related to location-dependent dendritic propagation and somatic EPSP normalization [15]. Active state, NMDA voltage dependence, clustered input, and timing can change voltage trajectories without changing the morphology-derived ratio. Equal SMI cannot imply electrical equivalence because many circuits share a ratio while differing in absolute RC constants, impedance phase, conductance drive, and morphology.

This does not make the ratio useless. It makes it conditional. A scalar descriptor can be valuable when its target is explicit and its residuals are measured. It becomes misleading only when the local divider coordinate is stretched into an amplitude, somatic, nonlinear, or categorical claim.

The radius audit highlights a measurement boundary. Because $R_{\text{neck}} \propto 1/r_n^2$, modest radius error can produce large resistance and ratio error, and spine-neck geometry is itself plastic and difficult to measure precisely [16, 17]. The N=768 radius screen produced a 24.0% design-fraction class flip, with all 11 intermediate-SMI combinations crossing a class boundary. This supports continuous reporting with explicit uncertainty more strongly than thresholded low/intermediate/high labels.

The same N=768 ensemble contained no high-SMI rows. That absence is a limitation, not a result to smooth over. High-isolation behavior is supported in this manuscript by reference cases and designed challenge rows, whereas uncertainty stability is supported only for sampled low/intermediate combinations. A future high-SMI uncertainty extension would need predefined ranges that actually sample that regime.

The from-scratch passive implementation is computationally credible within the scope tested. Internal unit, current-balance, limiting-case, timestep, impedance, and BE-versus-CN checks support numerical self-consistency. More importantly for the reviewer critique, Phase 4 added an independent direct-matrix implementation that reproduced baseline passive traces and metrics to roundoff, plus DC analytic benchmarks for the dendrite-soma load and divider limits.

This is still not the same as validation against NEURON or another community simulator. NEURON was unavailable in the execution environment and is not claimed. The strongest supported statement is narrower: the passive three-compartment reference equations are implemented consistently with an independently assembled direct matrix benchmark and the analytic DC limits. Extending that level of independent validation to distributed morphologies, active protocols, and a future optional NEURON benchmark remains useful future work.

The results fit naturally with mixed experimental evidence about spine electrical isolation. Some optical and uncaging studies support substantial compartmentalization in particular contexts, whereas other work emphasizes modest isolation, indicator limits, and context-dependent interpretation [5, 7, 9, 13, 14]. A manuscript that reports only a single ratio would flatten these differences.

A manuscript that reports the ratio, divider expectation, residuals, and target-specific descriptors can connect morphology to physiology without overclaiming.

The current framework remains intentionally limited. The baseline soma is a lumped downstream load, passive morphologies are procedural rather than cell-specific reconstructions, active conductances are generic stress tests rather than calibrated densities, and uncertainty ensembles are deterministic sensitivity designs rather than posterior biological distributions. SPINE focuses on electrical behavior and does not model biochemical calcium reaction-diffusion, receptor trafficking, signaling cascades, or structural plasticity.

The most valuable next studies would combine voltage imaging, morphology measurements, and electrophysiology in preparations where neck-radius uncertainty can be propagated explicitly. Modeling extensions should prioritize calibrated cell-type active conductances, morphology-diverse posterior inference, predefined high-SMI uncertainty coverage, and optional community-simulator cross-validation. Until then, the safest scientific claim is the one this rewrite foregrounds: the divider is expected; the residuals are informative.

The load-normalized spine-neck ratio is therefore a compact coordinate for the classical local voltage divider, not a universal electrical phenotype. Its most defensible use is to state the expected low-frequency head-to-dendrite transfer and then measure the residual from that expectation. In SPINE, those residuals show when transient conductance dynamics, impedance, synaptic drive, active state, morphology, downstream filtering, and measurement uncertainty make the scalar descriptor incomplete.

3 Online Methods

3.1 Reference circuit and load-normalized ratio

SPINE was implemented in Python using project-local equations, matrix assembly, solvers, metrics, and validation scripts. General scientific Python libraries are used for numerical linear algebra and analysis, but no prebuilt neuroscience simulator or copied model code is used as the runtime foundation. Baseline, plausibility-revised, active-extension, and exploratory configurations are separated so that extensions are not silently folded into the baseline reference track.

The baseline reference model contains a spine head h , a parent dendrite compartment d , and a soma/load compartment s . For a cylindrical spine neck with length L_n , radius r_n , and intracellular resistivity ρ_i ,

$$R_{\text{neck}} = \frac{\rho_i L_n}{\pi r_n^2}, \quad (3)$$

with geometry converted to centimeters before resistance is computed. The baseline target neck geometries are listed in Table 1. The head area is approximated as $4\pi r_h^2$ and sets head capacitance and leak; dendrite and soma capacitances and leaks are lumped. The soma is therefore a compact downstream load rather than a reconstructed full neuron.

The dendrite-soma input resistance $R_{\text{in},d}$ is measured after removing the stimulated spine and neck edge. A small current perturbation is applied to the dendrite-soma subcircuit, and $\Delta V_d/\Delta I_d$ is computed from the steady-state passive linear system with a perturbation simulation check. We then define the load-normalized spine-neck ratio

$$\text{SMI} = \frac{R_{\text{neck}}}{R_{\text{in},d}}. \quad (4)$$

This spine-omitted definition avoids normalizing a neck by a load that already contains the same neck. Phase 2 and Phase 4 one-port reconstructions showed that attached-versus-omitted $R_{\text{in},d}$

differences were numerically negligible for the compact passive-head model family, but the omitted definition remains conceptually cleaner.

3.2 Analytic divider and residual target

The analytic local reference is the DC small-signal voltage divider:

$$\Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + \text{SMI}}. \quad (5)$$

This equation is treated as classical circuit logic, not as a novel result. The manuscript's main diagnostic quantity is the residual between the simulated peak local transfer and this divider expectation:

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (6)$$

Here $\Gamma_{h \rightarrow d} = A_d/A_h$ is measured from transient conductance-based simulations over the predefined metric window. Because Γ_{div} is a DC low-frequency reference and $\Gamma_{h \rightarrow d}$ is a peak transient ratio, nonzero residuals are expected whenever capacitance, changing synaptic driving force, morphology, impedance, active state, or downstream coupling matters.

3.3 Passive equations and numerical solution

Passive voltages obey conductance-balance equations for leak, synaptic, head-dendrite, and dendrite-soma terms:

$$C_h \frac{dV_h}{dt} = -g_{L,h}(V_h - E_L) - g_{\text{syn}}(t)(V_h - E_{\text{syn}}) + g_{hd}(V_d - V_h), \quad (7)$$

$$C_d \frac{dV_d}{dt} = -g_{L,d}(V_d - E_L) + g_{hd}(V_h - V_d) + g_{ds}(V_s - V_d), \quad (8)$$

$$C_s \frac{dV_s}{dt} = -g_{L,s}(V_s - E_L) + g_{ds}(V_d - V_s). \quad (9)$$

Letting $\mathbf{V} = (V_h, V_d, V_s)^T$ and $\mathbf{C} = \text{diag}(C_h, C_d, C_s)$ gives

$$\mathbf{C} \frac{d\mathbf{V}}{dt} = -\mathbf{G}(t)\mathbf{V} + \mathbf{b}(t), \quad (10)$$

where

$$\mathbf{G}(t) = \begin{bmatrix} g_{L,h} + g_{\text{syn}}(t) + g_{hd} & -g_{hd} & 0 \\ -g_{hd} & g_{L,d} + g_{hd} + g_{ds} & -g_{ds} \\ 0 & -g_{ds} & g_{L,s} + g_{ds} \end{bmatrix}. \quad (11)$$

Backward Euler advances the passive state by solving

$$\left(\frac{\mathbf{C}}{\Delta t} + \mathbf{G}_{n+1} \right) \mathbf{V}_{n+1} = \frac{\mathbf{C}}{\Delta t} \mathbf{V}_n + \mathbf{b}_{n+1}. \quad (12)$$

Crank-Nicolson comparisons are used as internal numerical self-consistency checks, not as independent external validation.

The synapse is conductance based:

$$I_{\text{syn},h}(t, V_h) = g_{\text{syn}}(t)(V_h - E_{\text{syn}}), \quad (13)$$

where $g_{\text{syn}}(t)$ is a normalized double exponential with peak conductance g_{max} . Peak depolarizations A_h , A_d , and A_s are measured over a predefined 50 ms post-event window. Attenuation ratios are $\Gamma_{h \rightarrow d} = A_d/A_h$ and $\Gamma_{h \rightarrow s} = A_s/A_h$. $\Gamma_{h \rightarrow d}$ measures local transfer across the neck, $\Gamma_{h \rightarrow s}$ combines local transfer with downstream dendrite-to-soma filtering, and A_h measures local voltage excursion.

3.4 Challenge suites and descriptor comparisons

Passive morphology analyses generalize the reference circuit to compartment trees assembled from nodes and axial edges. Frequency-domain analyses solve the linearized passive system to estimate input impedance, transfer impedance, gain, phase, and dynamic SMI variants. Active and NMDA mechanisms are disabled in the baseline reference configuration and enabled only in active-extension protocols. These active conductances are generic stress tests using literature-traceable current forms [18–24], not calibrated cell-type-specific channel densities.

Descriptor comparisons were performed on existing derived rows. We compared raw SMI, the analytic divider transform Γ_{div} , component resistances R_{neck} and $R_{\text{in},d}$, log-component models, impedance and dynamic descriptors, transfer gain, and synaptic conductance scale where available. These comparisons were interpreted by target and descriptor family. Exact winner labels were avoided when near-tied or correlated predictors made the ranking unstable.

Designed counterexamples tested logical sufficiency. Iso-SMI cases ask whether equal ratio implies equivalent voltage responses; iso-neck-resistance cases ask whether equal R_{neck} can change under different loads; iso-transfer cases ask whether similar transfer can arise from different ratios. These experiments were designed to expose limitations as well as successes.

3.5 Deterministic uncertainty and reporting language

Uncertainty analyses used deterministic Latin-hypercube ensembles and progressive sample sizes of $N=96, 192, 384$, and 768 . The global uncertainty seed was 202605 ; progressive ensembles used $202605 + 5100 + N$. Radius-only uncertainty was analyzed separately because $R_{\text{neck}} \propto 1/r_n^2$, using 25 deterministic offsets spanning ± 2 SD with $\text{SD} = 0.018 \mu\text{m}$ and clipping measured radius to $0.06\text{--}0.22 \mu\text{m}$. Heuristic SMI classes were low < 0.25 , intermediate 0.25 to < 0.75 , and high ≥ 0.75 .

All uncertainty fractions are reported as deterministic design fractions over sampled parameter combinations. They are not biological rates and are not treated as inferential population intervals. The $N=768$ ensemble contained 757 low-SMI rows, 11 intermediate-SMI rows, and zero high-SMI rows, so high-SMI uncertainty claims are explicitly scoped to reference and designed challenge rows rather than to the main uncertainty ensemble.

3.6 Computational credibility checks

Numerical credibility was assessed at three levels. First, internal checks covered units, current signs, synaptic normalization, limiting cases, timestep refinement, sparse/dense passive assembly, impedance validation, and Backward Euler versus Crank-Nicolson peak differences. Second, Phase 4 added an independent direct-matrix benchmark that separately assembled the baseline passive three-compartment equations and compared against existing SPINE trace outputs without calling the production passive simulation function as its benchmark engine. Third, DC analytic checks tested the two-node dendrite-soma input resistance, divider formula, limiting cases, and attached-versus-omitted one-port behavior.

NEURON availability was checked during Phase 4. It was not available in the bundled runtime, no installation was attempted, and no NEURON result is claimed. The executed validation therefore supports the passive reference implementation through independent direct-matrix and analytic benchmarks, not through a community-simulator cross-validation.

3.7 Traceability

All manuscript-facing values are rounded for readability while exact values remain in machine-readable CSV outputs and ledgers. The claim-to-source ledger links major quantitative statements to source reports and CSV rows, following model-sharing and FAIR-data principles [25, 26]. Figure and table manifests record provenance for publication assets. Exploratory perturbation scenarios remain supplemental stress tests and are not used to support primary biological mechanism claims.

Disclosures

Funding: This work received no external funding.

Competing interests: The authors declare no competing interests.

Ethics statement: This manuscript reports mathematical and computational modeling using repository simulations and no new human participants or animal experiments.

Author contributions: Gregory Pierpoint: Methodology; Software; Formal analysis; Investigation; Validation; Visualization; Data curation; Writing – original draft; Writing – review & editing. Alberto Musto: Conceptualization; Methodology; Supervision; Project administration; Writing – review & editing; Scientific interpretation; Final manuscript approval.

AI-use disclosure: Generative-AI tools were used for editorial organization, code-development assistance, revision planning, and manuscript-formatting support. The authors retain responsibility for study conception, model interpretation, validation, numerical verification, reference verification, and final manuscript content. No generative-AI tool is listed as an author.

Data Availability

This study uses mathematical and computational modeling and generated no new human participant or animal data. Manuscript-facing derived data, validation outputs, figure/table source data, and provenance ledgers are included in the SPINE release package. Public release is approved. A release-ready repository package has been prepared for GitHub release as `spine-divider-residuals`; final GitHub URL, release commit, and Zenodo DOI require manual authenticated upload/integration because the local environment did not provide an authenticated GitHub CLI or Zenodo token.

Manuscript claim provenance is recorded in `manuscript/CLAIM_TO_SOURCE_LEDGER.csv`. Figure and table provenance are recorded in `manuscript/FIGURE_SOURCE_MANIFEST.csv` and `manuscript/TABLE_SOURCE_MANIFEST.csv`. These ledgers are traceability aids and do not replace the manuscript’s scoped claims.

Code Availability

The SPINE Python code, configuration files, analysis scripts, tests, validation scripts, environment instructions, and release manifests are included in the approved public-release staging package. Code is released under the MIT License. Derived manuscript-facing data, figures, tables, and documentation are released under CC BY 4.0 unless otherwise stated. Final public GitHub URL, release tag/commit hash, and Zenodo DOI should be inserted after authenticated manual upload and archival integration.

References

- [1] Wilfrid Rall. Theoretical significance of dendritic trees for neuronal input-output relations. In *Neural Theory and Modeling*, pages 73–97. Stanford University Press, Stanford, CA, 1964. doi: 10.7551/mitpress/6743.003.0015.
- [2] Idan Segev and Wilfrid Rall. Computational study of an excitable dendritic spine. *Journal of Neurophysiology*, 60(2):499–523, 1988. doi: 10.1152/jn.1988.60.2.499.
- [3] David Tsay and Rafael Yuste. On the electrical function of dendritic spines. *Trends in Neurosciences*, 27(2):77–83, 2004. doi: 10.1016/j.tins.2003.11.008.
- [4] Rafael Yuste. Electrical compartmentalization in dendritic spines. *Annual Review of Neuroscience*, 36:429–449, 2013. doi: 10.1146/annurev-neuro-062111-150455.
- [5] Roberto Araya, Jiang Jiang, Kenneth B. Eisenthal, and Rafael Yuste. The spine neck filters membrane potentials. *Proceedings of the National Academy of Sciences of the United States of America*, 103(47):17961–17966, 2006. doi: 10.1073/pnas.0608755103.
- [6] Mark T. Harnett, Judit K. Makara, Nelson Spruston, William L. Kath, and Jeffrey C. Magee. Synaptic amplification by dendritic spines enhances input cooperativity. *Nature*, 491(7425):599–602, 2012. doi: 10.1038/nature11554.
- [7] Taekyung Kwon, Miki Sakamoto, Darcy S. Peterka, and Rafael Yuste. Attenuation of synaptic potentials in dendritic spines. *Cell Reports*, 20(5):1100–1110, 2017. doi: 10.1016/j.celrep.2017.07.012.
- [8] L. Beaulieu-Laroche and Mark T. Harnett. Dendritic spines prevent synaptic voltage clamp. *Neuron*, 97(1):75–82.e3, 2018. doi: 10.1016/j.neuron.2017.11.016.
- [9] Victor Hugo Cornejo, Netanel Ofer, and Rafael Yuste. Voltage compartmentalization in dendritic spines in vivo. *Science*, 375(6576):82–86, 2022. doi: 10.1126/science.abg0501.
- [10] Jeffrey C. Magee. Dendritic integration of excitatory synaptic input. *Nature Reviews Neuroscience*, 1(3):181–190, 2000. doi: 10.1038/35044552.
- [11] Michael London and Michael Hausser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. doi: 10.1146/annurev.neuro.28.061604.135703.
- [12] Greg J. Stuart and Nelson Spruston. Dendritic integration: 60 years of progress. *Nature Neuroscience*, 18(12):1713–1721, 2015. doi: 10.1038/nn.4157.
- [13] Marko A. Popovic, Nicholas Carnevale, Balazs Rozsa, and Dejan Zecevic. Electrical behaviour of dendritic spines as revealed by voltage imaging. *Nature Communications*, 6:8436, 2015. doi: 10.1038/ncomms9436.
- [14] Dejan Zecevic. Electrical properties of dendritic spines. *Biophysical Journal*, 122(22):4303–4315, 2023. doi: 10.1016/j.bpj.2023.10.008.
- [15] Jeffrey C. Magee and Erik P. Cook. Somatic epsp amplitude is independent of synapse location in hippocampal pyramidal neurons. *Nature Neuroscience*, 3(9):895–903, 2000. doi: 10.1038/78800.

- [16] Jan Tønnesen, Gergely Katona, Balázs Rózsa, and U. Valentin Nägerl. Spine neck plasticity regulates compartmentalization of synapses. *Nature Neuroscience*, 17(5):678–685, 2014. doi: 10.1038/nn.3682.
- [17] Jan Tønnesen and U. Valentin Nägerl. Dendritic spines as tunable regulators of synaptic signals. *Frontiers in Psychiatry*, 7:101, 2016. doi: 10.3389/fpsy.2016.00101.
- [18] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4): 500–544, 1952. doi: 10.1113/jphysiol.1952.sp004764.
- [19] Mark L. Mayer, Gary L. Westbrook, and Peter B. Guthrie. Voltage-dependent block by mg²⁺ of nmda responses in spinal cord neurones. *Nature*, 309:261–263, 1984. doi: 10.1038/309261a0.
- [20] Craig E. Jahr and Charles F. Stevens. A quantitative description of nmda receptor-channel kinetic behavior. *The Journal of Neuroscience*, 10(6):1830–1837, 1990. doi: 10.1523/JNEUROSCI.10-06-01830.1990.
- [21] Jeffrey C. Magee. Dendritic hyperpolarization-activated currents modify the integrative properties of hippocampal ca1 pyramidal neurons. *The Journal of Neuroscience*, 18(19):7613–7624, 1998. doi: 10.1523/JNEUROSCI.18-19-07613.1998.
- [22] Jeffrey C. Magee. Dendritic ih normalizes temporal summation in hippocampal ca1 neurons. *Nature Neuroscience*, 2(6):508–514, 1999. doi: 10.1038/9158.
- [23] John A. Connor and Charles F. Stevens. Prediction of repetitive firing behaviour from voltage clamp data on an isolated neurone soma. *The Journal of Physiology*, 213(1):31–53, 1971. doi: 10.1113/jphysiol.1971.sp009366.
- [24] John R. Huguenard and David A. McCormick. Simulation of the currents involved in rhythmic oscillations in thalamic relay neurons. *Journal of Neurophysiology*, 68(4):1373–1383, 1992. doi: 10.1152/jn.1992.68.4.1373.
- [25] Michael L. Hines, Thomas Morse, Michele Migliore, Nicholas T. Carnevale, and Gordon M. Shepherd. Modeldb: A database to support computational neuroscience. *Journal of Computational Neuroscience*, 17(1):7–11, 2004. doi: 10.1023/B:JCNS.0000023869.22017.2e.
- [26] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J. G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A. C. 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. The fair guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1): 160018, 2016. doi: 10.1038/sdata.2016.18.